Implicit Q-Learning on HalfCheetah

RL project proposal

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IQL: learning from a fixed dataset

Classification	Offline, model-free actor-critic RL
Training data	Recorded transitions: state, action, reward, next state
Main idea	Learn values from dataset actions, then give better actions more weight when fitting the policy
Interaction	Simulator rollouts evaluate the policy after training

The IQL update

State value V	Upper expectile regression on dataset Q-values Expectile = 0.7
Twin critics Q1, Q2	Target = reward + 0.99 × (1 - terminated) × V(next state)
Policy	Advantage = min(target Q1, target Q2) - V Weighted log-likelihood of recorded actions
Policy weight	min(exp(3 × advantage), 100) Tanh-squashed Gaussian actions stay within [-1, 1]

HalfCheetah-v5: observation and action

Environment

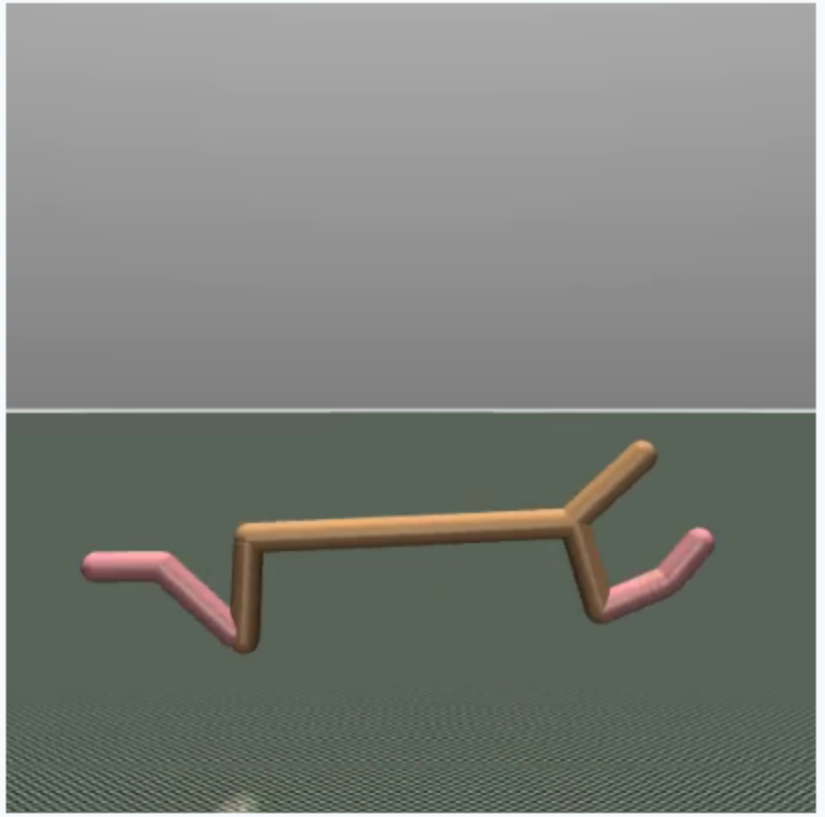
2D locomotion in Gymnasium and MuJoCo

Observation: 17 values

8 position / angle values + 9 velocities
Absolute horizontal position is excluded

Action: 6 values

Continuous torques for front and back leg joints
Each action component lies in [-1, 1]



Reward, goal and episode ending

Reward	Forward velocity - 0.1 × sum(action²)
Agent goal	Sustained forward motion with efficient torque use
Episode ending	No natural termination, including after a fall Time limit: 1,000 steps, about 50 simulated seconds
Learning target	A coordinated gait that maintains speed Time-limit truncation preserves value bootstrapping

Offline data and evaluation protocol

Dataset	Minari mujoco/halfcheetah/medium-v0 1,000 episodes and 1,000,000 transitions
Training budget	500,000 offline updates per training seed Seeds 0, 1 and 2, batch size 256
Evaluation	Intermediate checkpoints, 10 episodes per run Raw return, uniform random baseline and full rollouts
Reproducibility	Dataset-matched environment, MuJoCo 3.2.3 Separate checkpoint selection and final test episodes

Why IQL and HalfCheetah fit

Continuous control	The policy learns bounded joint torques directly
Available offline data	A fixed locomotion dataset supports IQL training without collecting new training rollouts
Learning opportunity	Advantage weights emphasize better recorded actions Improvement over the behavior policy is a hypothesis
Practical environment	Standard benchmark with measurable returns and visible locomotion behavior

Challenges and final-project plan

Coverage and stability	Limited data coverage and seed-dependent failures Multiple seeds, separate final tests and full videos
Reliable comparisons	Physics versions and checkpoint selection affect results Pinned dependencies and documented evaluation seeds
Team ownership	Aleksandr: integration and reproducibility Nikita: environment and dataset; Leo: IQL Ruslan: evaluation; Telman: notebook and presentation
Final deliverables	Reproducible Colab, checkpoints, evaluation plots and a complete policy demonstration